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GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	**Green Initiatives**
Document Number	EA-BFPCL-GPMC-GGPL-FEED-ELE-RPT-005
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Green Initiatives

R01	09.08.22	Issued for Review	N. Kari	V. Ani	M.Ekpunobi	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	09.08.22	Issued for Review

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.

b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)

for the subsequent project phases, post FEED, it is expected that:

c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.

d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)

e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED-ELE-RPT-005

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviation	Meaning
AC	Alternating Current
DB	Distribution Board
DC	Direct Current
EPC	Engineering, Procurement and Construction
EMC	Electromagnetic Compatibility
IEC	International Electro Technical Commission
IEEE	Institute of Electrical and Electronics Engineers
IFC	Issued for Construction
IFD	Issued for Design
IP	Ingress Protection
ISO	International Organisation for Standardization
LV	Low Voltage
MTO	Materials Take Off
PCS	Process Control System
PFD	Process Flow Diagram
P&ID	Piping and Instrumentation Diagram
UPS	Uninterruptible Power Supply
XPLE	Cross-Linked Polyethylene

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

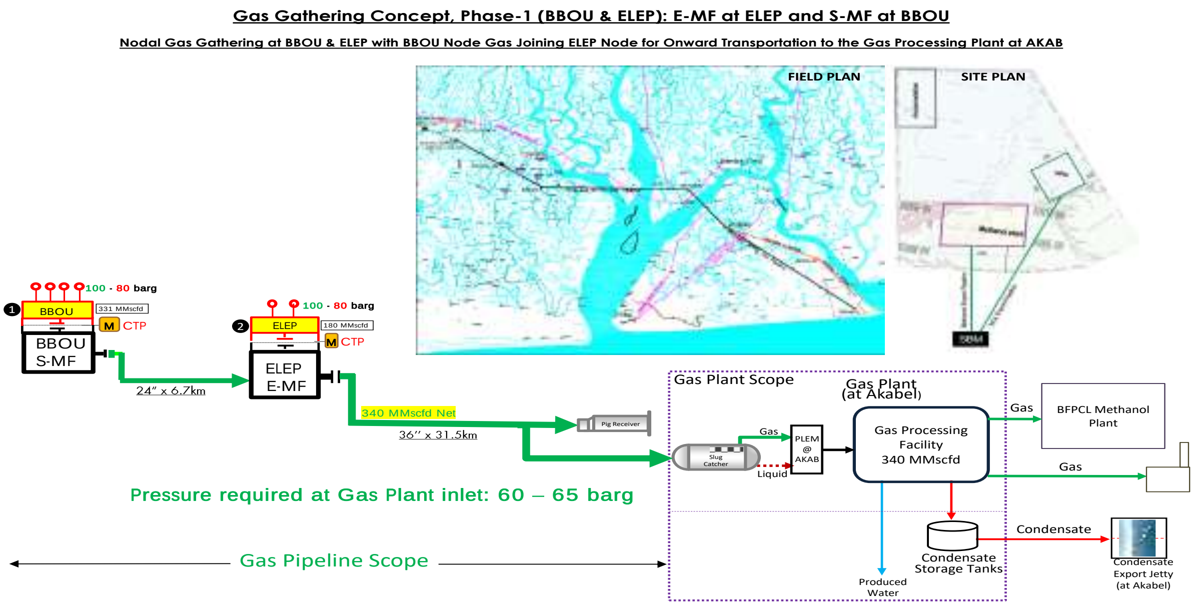

Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The scope of this report is to define the project approach towards Green initiatives aimed at reducing emissions into the atmosphere.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
IEC 62109-2	Covers the particular safety requirements relevant to d.c. to a.c. inverter products
IEC 60947	Low-voltage switchgear and control gear
IEC 60269	Low-voltage fuses
IEC 60947-2	Low-voltage switchgear and Control gear. Part 2: Circuit breakers
IEC 60947-3	Low voltage switchgear and control gear - Part 3: Switches, disconnectors, switch-disconnectors and fuse-combination units.
IEC 60947-4-1	Low voltage switchgear and control gear - Part 4: Contactors and motor starters - Section one - Electromechanical contactors and motor starters.
IEC 60364-1	Low voltage electrical installations; part 1; gen characteristics
IEC 60529	Classification of Degrees of Protection Provided by Enclosures
IEC 62053-11	Electrical metering Equipment; Class 0.5, 1 and 2 alternating-current watt-hour meters.
IEC 60304	Standard Colours for Insulation for Low-Frequency Cables and Wire
IEC 60721-2-1	Classification of Environmental Conditions: Part 2-1:
IEC 61554	Dimensions for panel-mounted indicating and recording electrical
IEC 61508	Safe Capability: system related with safety
IEC 60079	Electrical apparatus for explosive gas atmospheres (part 10 & 14)
IEC 60099	Surge arresters and control gear standards
IEC 60073	Basic and safety principles for man-machine interface, marking and identification - Coding principles for indicators and actuator
IEC 62353	Electrical safety testing
ISO 9001:2000	Quality Management Systems-Requirements
IEEE C37.2	ANSI/IEEE Standard Device Numbers

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Executive Summary

The aim of Green Initiative is to reduce the project Green House Gas (GHG) emissions and optimize energy efficiency.

The emissions and energy consumption of the Gas Gathering Pipelines and Associated Infrastructure are minimal as there is no facility scope. Despite this, concerted effort has been made to reduce GHG emissions within the system to As Low As Reasonably Practicable.

Summary of principal measures adopted within the process to avoid / reduce GHG emissions include;

1. No flaring or continuous venting of hydrocarbons during normal operations.
2. Use of gas-over-oil system for actuated valves instead of instrument air thereby eliminating the need for electricity consuming instrument air compressors.
3. Non-use of electricity consuming motor-operated valves.
4. Reduction of potential fugitive release sources. Fugitive releases are those that occur routinely at extremely low flow rates from process equipment such as valves and flanges. Welded connections have been preferentially selected over flanged connections where practicable.

4. Electric Power Supply System

A significant portion of emissions and energy consumption is usually from Electric Power generation hence particular attention has been paid to the selection of the power supply and distribution system.

4.1. Primary Power Supply System

Electric power will be supplied from the power plant in the Methanol Complex at Akabel through underground submarine cable. This will reduce the foot print of GHG emissions as there will be no direct GHG emission on site, and being a very reliable power supply system that requires no back-up, will eliminate the need for GHG-emitting back up fossil fuel electric generators.

4.2. Renewable Energy Power System

The ideal Green Initiative electric power supply system is renewable energy like solar, wind or bio fuel. Unfortunately, there isn't enough data available to support the deployment of any of these systems on the project. A pilot solar power system will however be installed at BBOU to serve as a test case for the viability of solar power system in the area. If proven viable, solar power system will be deployed in the future locations of NUNR, SEIB and OPUG, and may be retrofitted at the current locations of BBOU and ELEP if proven economical, and/or becomes required by law.

4.3. Off Grid Solar Inverter System

An off-grid solar inverter system shall be designed to provide total power requirement of the Manifold Stations for the implementation of green initiatives.

The inverter system will be designed to guaranty enough solar panels to charge a dual redundant battery system that will support minimum of 3 days autonomy. Off-grid solar inverter system will involve the use of the following systems.

- Installation of chargers, batteries and inverters.
- Installation of PV Panels in a non-classified location.

4.3.1. Merits

- This is a renewable Energy Supply System that will be most welcome in the current campaign to reduce the use of fossil fuels and the attendant negative impact on the environment.
- Less expensive when compared with Underground Cable Power.
- Is Most suitable for unmanned remote manifold station.
- Minimum maintenance requirement.